

Question\_4:

Code:

```
1  /*
2  Question No: 4
3  Given a weighted graph, find the total weight of its
4  maximum spanning tree.
5  Take edges from largest weight to smallest and use DSU.
6  */
7  #include <iostream>
8  #include <vector>
9  #include <algorithm>
10 using namespace std;
11
12 struct Edge{
13     int u,v,w;
14 };
15 int p[5005];
16 int find(int x){
17     return p[x]==x?x:p[x]=find(p[x]);
18 }
19 int printheap(int N){
20     return N;
21 }
22 int main(){
23     int t;
24     cin>>t;
25     while(t--){
26         int n,m;
27         cin>>n>>m;
28         vector<Edge> e(m);
29         for(auto &x:e)
30             cin>>x.u>>x.v>>x.w;
31         sort(e.begin(),e.end(),[](Edge a,Edge b){
32             return a.w>b.w;
33         });
34         for(int i=1;i<=n;i++)
35             p[i]=i;
36         long long ans=0;
37         for(auto x:e){
38             int a=find(x.u),b=find(x.v);
39
40             if(a!=b){
41                 p[a]=b;
42                 ans+=x.w;
43             }
44         }
45         cout<<ans<<"\n";
46     }
47 }
```

**Output:**

**Test Case\_1:**

```
1
3 3
1 2 2
2 3 3
1 3 4
7
PS C:\Users\Prabin Yadav\
```

**Test Case\_2:**

```
1
3 3
1 2 3
2 3 4
1 3 5
9
PS C:\Users\Prabin Yadav\
```